

Trends in Female Breast Cancer Incidence among Japanese, Korean, and US Populations: An Age-Period-Cohort Analysis

Aminu Kende Abubakar^{1,2}, Hasan Jamil^{1,2}, Hirokazu Tanaka¹, and Kota Katanoda¹



ABSTRACT

Background: Breast cancer incidence has historically been higher in Western countries; however, rates have increased substantially in East Asia, with variation by age and generation. Direct comparisons between native Asians and their US counterparts remain limited. We examined the effects of age-period-cohort (APC) on breast cancer incidence among native Japanese women, with comparisons with Japanese Americans, native Korean women, Korean Americans, and US White and Black women.

Methods: We analyzed population-based cancer registry data from Japan (1985–2019), Korea (1993–2017), Hawaii (1988–2017), Los Angeles County (1988–2017), and Surveillance, Epidemiology, and End Results 8 (1985–2019). Women 25 to 84 years of age were grouped into 5-year age groups and calendar periods, and APC analyses were conducted using the NCI APC web-based tool.

Results: Incidence increased with age in all populations; however, the patterns differed. Native Japanese and Koreans had the lowest overall rates with an early-midlife peak, whereas Japanese Americans resembled US Whites, with incidence increasing into older ages. Period and cohort effects were strongest among native Japanese and Koreans, with sustained increases among women born after 1950. Annual incidence increases were highest among native Koreans (5.3%) and native Japanese (3.8%), followed by Korean Americans (2.8%) and Japanese Americans (1.6%), whereas US Whites were stable or declining and US Blacks showed modest increases.

Conclusions: These findings indicate an ongoing epidemiologic transition in East Asia and demonstrate how migration and generational change shape breast cancer risk across populations.

Impact: Direct comparison of native and US Asian populations reveals migration-related divergence in breast cancer risk.

Introduction

Female breast cancer is a leading cause of cancer morbidity and mortality worldwide (1). In Japan, it is the most commonly diagnosed cancer and the fourth leading cause of cancer deaths among women, with 99,449 cases reported in 2021 and 15,629 deaths in 2023 (2). Globally, incidence and mortality have increased, and Japan has followed a similar trend (1, 3). Incidence reflects changing risk-factor profiles and health-system factors, including screening and diagnostic intensity. Historically, Western countries have reported higher incidence rates than Asian countries (4), but rates have increased in several Asian settings, including Japan and Korea, and among Asian women in the United States (1, 5) while stabilizing or declining in many Western populations, particularly among US White women (5). These opposing trends have been attributed to changes in risk-factor profiles, screening practices, and advances in diagnostic technologies (6–8).

Breast cancer risk and outcomes vary across age groups and generations. For example, age-specific incidence patterns differ. Western women observe a rapid increase until menopause, followed by a slower increase, whereas Asian women typically plateau or

decline after menopause. This contrast, noted in prior studies, is debated, with explanations including differences in cohort dynamics, identification, and risk-factor settings (9–11). Moreover, risk factors such as obesity, weight gain, and physical activity may affect age groups and racial or ethnic groups differently (12, 13). Because calendar-period shifts and birth-cohort replacement can both change observed rates, separating age, period, and cohort patterns is important for interpreting Japan's increasing burden and for setting prevention and screening priorities (3, 14). Migration studies further support the role of modifiable environmental and lifestyle factors, as Asian women in Western countries often encounter host-population risk (15). These differences highlight limitations of traditional trend analyses and the need for a more refined approach, considering biological, temporal, and generational influences.

Age-period-cohort (APC) analysis offers a useful framework for understanding how age, calendar time, and generational change jointly shape the epidemiology of chronic diseases over time (16). Age effects reflect biological aging; period effects capture external influences such as screening and medical advances; and cohort effects highlight shared risks among birth groups. Although APC models have been used to study breast cancer trends in Japan and Asia, most Japanese studies rely on data from limited local registries and on older datasets obtained before 2000 (17, 18). More recent studies have focused primarily on global and regional comparisons without examining population-specific patterns in detail (19–21). Comparisons between Asian and Western groups often exclude US-based Asians, and US studies typically aggregate diverse Asian populations into a single group category (10, 22). Only 1 prior study has systematically compared native Asian and US-based Asian populations, and similarly it relied on pre-2000 data (11). Current evidence is insufficient to clarify how generational and temporal factors influence breast cancer rates among native Asian women and Asian Americans in today era. To address this gap, we compared

¹Division of Population Data Science, National Cancer Center Institute for Cancer Control, Tokyo, Japan. ²Graduate School of Public Health, St. Luke's International University, Tokyo, Japan.

Corresponding Author: Aminu Kende Abubakar, Division of Population Data Science, National Cancer Center Institute for Cancer Control, Tokyo, Japan, 5 Chome-1-1 Tsukiji, Chuo City, Tokyo 104-0045, Japan. E-mail: aminukende8@gmail.com

Cancer Epidemiol Biomarkers Prev 2026;35:1120–8

doi: 10.1158/1055-9965.EPI-26-0101

This open access article is distributed under the Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International (CC BY-NC-ND 4.0) license.

©2026 The Authors; Published by the American Association for Cancer Research

Japanese women with US White and Black women as reference populations for established US incidence patterns and with US Asian groups to evaluate how incidence in Asian-origin populations is observed within a historically higher-incidence setting. We also included Korea as a nonmigrant East Asian reference to help distinguish region-wide temporal patterns from those that are more specific to Japan. Age, period, and cohort effects on breast cancer incidence were examined among native Japanese women, with comparisons with Japanese Americans, native Korean women, Korean Americans, and US White and Black women.

Materials and Methods

Data sources and study population

We analyzed female breast cancer data from 6 groups: native Japanese women in Japan, US Japanese women (Hawaii), US White and Black women, native Korean women in Korea, and US Korean women (California). For native Japanese women, data were obtained from the Cancer Information Service for cancer registries in Yamagata, Fukui, and Nagasaki prefectures, covering the period 1985 to 2019 (2). These registries were selected because Japan's national cancer registry was established in 2016, and the validity and national representativeness of these prefectural registries have been previously established (3). Although larger urban registries exist, they have limited or discontinuous coverage across age groups and periods, making them less suitable for an APC modeling study. Data for US Japanese, US Koreans, and native Koreans were sourced from the Hawaii, Los Angeles County, and Seoul registries, respectively, via Cancer Incidence in Five Continents (CI5, volumes VII–XII; RRID: SCR_027553). These registries offered the most stable long-term population-based coverage available for these subgroups. The estimates reflect the registry's catchment areas and may not fully represent the US Japanese and US Korean populations nationwide. Combined registries from Hawaii, Los Angeles, and San Francisco (volumes VII–XI) and national Korean data (volumes X–XII) were used in sensitivity analyses. These datasets were excluded from the primary analysis because they lacked continuous coverage for the US Japanese population and a shorter observation

period for native Koreans. US Asian groups were defined by registry-recorded race/ethnicity, not by nativity or migration status; thus, these categories include both US-born and foreign-born individuals (RRID: SCR_006902). Furthermore, the native Asian data do not include race/ethnicity, although a small fraction may include non-Japanese/Korean residents. Incidence data for US Whites and Blacks (1985–2019) were obtained from the Surveillance, Epidemiology, and End Results (SEER) Program, which is defined by the SEER race recode (White and Black), independent of Hispanic ethnicity (RRID: SCR_006902). Details of all databases are provided in **Table 1**.

Population denominators were obtained from each data source for age- and period-specific population counts (**Table 1**). The analysis included women 25 to 84 years of age, grouped into 5-year age categories and aggregated into 5-year calendar periods. We restricted the analyses to women 25 to 84 years of age to capture predominantly sporadic, adult-onset breast cancer. Cases were defined by International Classification of Diseases (10th Revision) code C50, excluding carcinoma *in situ*. Age-specific incidence rates were calculated within 5-year age groups and 5-year calendar periods.

All included registries met the International Agency for Research on Cancer (IARC) data quality criteria, including low death certificate–only registrations and high microscopically verified diagnoses. SEER registries also follow strict data quality standards and routinely evaluate completeness and validity. Given this quality assurance, we did not add registry-specific adjustments. This article was prepared following the Strengthening the Reporting of Observational studies in Epidemiology guidelines for reporting the findings of observational studies (<https://www.equator-network.org/reporting-guidelines/strobe/>). Ethical approval was waived as the study used deidentified, publicly available cancer registry data.

Statistical analysis

The APC analysis was conducted using the US National Cancer Institute (NCI) APC web tool (RRID: SCR_027554). This tool uses generalized linear models with weighted least squares to assess the independent effects of age (biological aging risk), period (calendar-time factors such as screening or treatment), and cohort (generational exposure and screening implementation) on breast cancer incidence

Table 1. Data sources and case counts for female breast cancer incidence by population group.

Population	Registry/source	Years of data	Population source	Total cases	Person-years at risk	Reference period	Reference cohort
Native Japanese	Cancer registries in Yamagata, Fukui, and Nagasaki prefectures	1985–2019	Japan National Vital Statistics	44,102	45,541,948	2000–2004	1950
US Japanese	USA, Hawaii (CI5 volumes VII–XII)	1988–2017	CI5	6,834	2,798,717	1998–2002	1948
	USA, California, Los Angeles County (CI5 volumes VII–XII)	1998–2007		1,689	1,015,576		
	USA, California, San Francisco Bay Area (CI5 volumes XII, IX, and XI)	1988–1992, 1998–2002, and 2008–2012		646	398,391		
US White	SEER 8	1985–2019	SEER/NCHS	410,332	203,369,316	2000–2004	1950
US Black	SEER 8	1985–2019	SEER/NCHS	37,558	24,751,377	2000–2004	1950
Native Koreans	Seoul Registry (CI5 volumes IX–XII)	1993–2017	CI5	66,144	70,861,774	1998–2002	1948
	All Korea (CI5 volumes IX–XII)	1999–2017		247,960	322,515,208		
US Koreans	USA, California, Los Angeles County (CI5 volumes VII–XII)	1988–2017	CI5	1,852	1,591,149	1998–2002	1948

NOTE: SEER 8, data extracted from the SEER 8 registry system, which includes Atlanta, San Francisco–Oakland, Connecticut, Detroit, Hawaii, Iowa, New Mexico, and Seattle–Puget Sound.

Abbreviation: NCHS, National Center for Health Statistics.

(23). The tool addresses the inherent nonidentifiability of APC models, which is due to the linear dependency among age, period, and cohort (cohort = period – age), by avoiding direct simultaneous estimation of all 3 effects. It provides estimable parameters, including longitudinal and cross-sectional age curves, period and cohort relative risks, net drift, and local drifts (23).

The longitudinal age curve represents expected age-specific incidence rates adjusted for period effects. In contrast, the cross-sectional age curve adjusts for cohort effects and reflects observed rates across different birth cohorts at specific calendar periods. Although both curves describe age-related patterns, the longitudinal curve more accurately captures biological aging, especially when substantial net drift is present (23). Period relative risks are rate ratios (RR) for 5-year calendar periods, calculated relative to reference periods that were selected based on the median years in each dataset (Table 1). Cohort relative risks were calculated for each birth cohort, relative to the median birth cohort in each dataset, as RRs (Table 1). Local drifts represent the annual percentage change in age-specific rates, whereas net drift indicates the overall annual percentage change in age-adjusted rates. Detailed formulas for all parameters used in this study are available in a previously published study (23). Statistical significance was assessed using the Wald χ^2 test, with *P* values < 0.05 considered significant. These tests assessed overall temporal trends (overall annual percentage change), deviations across calendar periods (period RRs), generational differences across birth cohorts (cohort RRs), and age-specific annual percentage change. All parameter estimates, including 95% confidence intervals (CI) and global Wald χ^2 test results, are provided in Supplementary Tables S1–S3.

Results

Study population

A total of 44,102 breast cancer cases were included for native Japanese women and 66,144 for native Korean women. Among the US populations, 6,834 cases were identified in the US Japanese women and

1,852 in the US Korean women. In addition, 410,332 cases were recorded for US White women and 37,558 for US Black women (Table 1).

Age-specific incidence

Figure 1 shows age-specific breast cancer incidence rates using longitudinal (Fig. 1A) and cross-sectional (Fig. 1B) curves. In the longitudinal curves, the incidence rate increased steadily with age across all groups, although the slope and peak levels varied. Native Japanese, native Koreans, and US Koreans consistently had the lowest rates across age groups, with native Japanese reaching 348.4 per 100,000 person-years among those of 80 to 84 years of age. In contrast, US Japanese, US White, and US Black women had higher incidence rates, with US Japanese women exceeding 500 per 100,000 person-years at older ages, whereas US White and Black women peaked around 420 to 450 per 100,000 person-years at older ages. In the cross-sectional curves, native Japanese and native Korean women exhibited the characteristic early-age peak around 45 to 49 years, followed by a plateau or slight decline. In contrast, the US Japanese women showed a distinct later trajectory, with incidence increasing steadily and peaking at 70 to 74 years, closely mimicking the pattern observed in US White women. US Korean women displayed an intermediate pattern between native Asian and US groups, with consistently lower rates than those of other US populations. Detailed age-specific incidence rates with corresponding 95% CI are provided in Supplementary Table S1.

Period effects

Figure 2 shows period effects in breast cancer incidence. Overall, native Japanese and native Korean women demonstrated the most pronounced and sustained increases across the study period. Among native Japanese women, the period RR increased to 1.88 (95% CI, 1.81–1.95) in 2015 to 2019, whereas among native Korean women it reached 2.11 (95% CI, 2–2.23) in 2013 to 2017. In contrast, the period RRs for the US Japanese women in the later periods were not

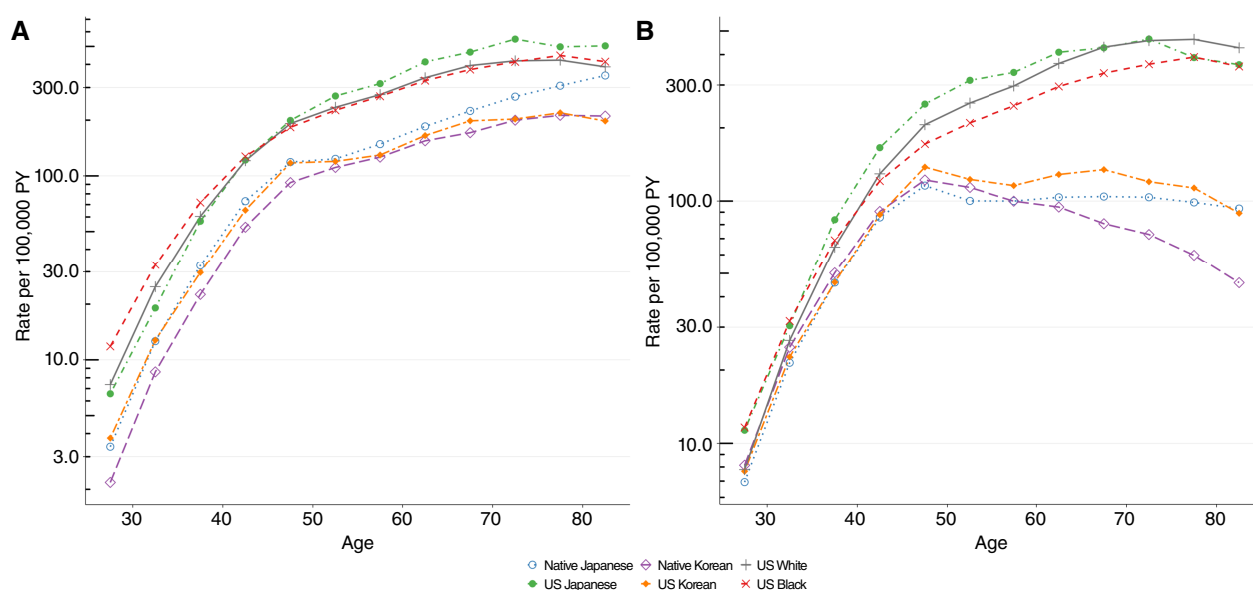


Figure 1.

Age-specific breast cancer incidence rates. **A**, Longitudinal age-specific incidence rates. **B**, Cross-sectional age-specific incidence rates. Rates are shown per 100,000 person-years (PY) and plotted at the midpoint of each 5-year age group.

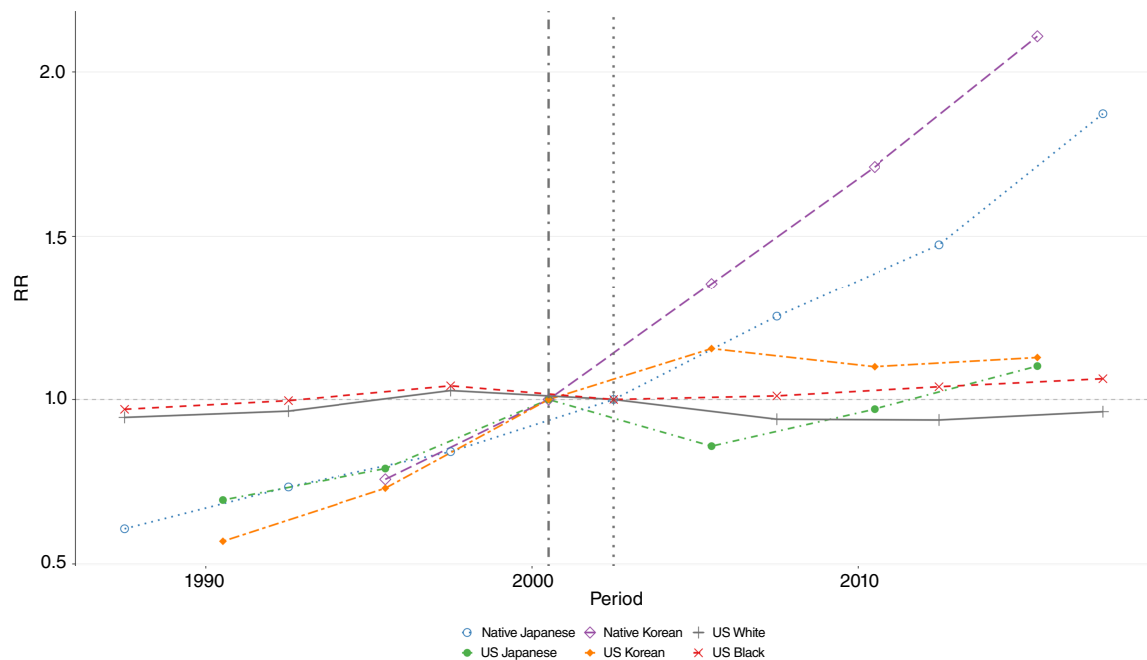


Figure 2.

Impact of period effects on breast cancer incidence. Period RRs by 5-year calendar periods, plotted at the midpoint of each period. The faint vertical dotted line marks the reference period (See **Table 1** for the reference period used for each population).

significantly different from those in the reference period. Among US racial groups, only US Black women experienced a modest increase, reaching 1.07 (95% CI, 1.03–1.11) in 2015 to 2019, whereas incidence in US White women remained stable across periods. Overall period effects were significant for all populations ($P < 0.001$; Supplementary Table S3). Detailed period RRs with corresponding 95% CIs are provided in Supplementary Table S2.

Cohort effects

Figure 3 shows cohort effects in breast cancer incidence. Most populations exhibited increasing cohort RRs across successive birth cohorts, except for US Black women, who showed stable patterns over time. Native Japanese and native Korean women demonstrated the most consistent generational increases. Among native Japanese women, all cohorts born after 1950 experienced significantly elevated risks, with the highest RR of 2.31 (95% CI, 1.48–3.59) observed in the most recent cohort. A similar pattern was observed among native Korean women, who reached a peak RR of 5.80 (95% CI, 4.24–7.92) in the most recent cohort. US Japanese and US Korean women also exhibited marked increases in cohort effects after 1950. However, the 1988 cohort of US Korean women showed an anomalous decrease with wide uncertainty (RR, 0.66; 95% CI, 0.07–7.14) that was not significant, possibly due to unstable estimates from sparse case counts. Overall cohort effects were significant for each population ($P < 0.001$). Detailed cohort RRs with corresponding 95% CI are provided in Supplementary Table S2.

Annual percentage change

Figure 4 presents the annual percentage change by age group (**Fig. 4A**) and overall (**Fig. 4B**). Native Japanese and native Korean women had the largest age-specific increase, peaking at 4.7% and 6.2%, respectively, in the 65 to 69 years age group (**Fig. 4A**). Similarly, US

Korean women showed a comparable age-related pattern, with slightly lower levels than those of native Koreans. In contrast, the US Japanese women had moderate increases across most age groups. Meanwhile, US White and US Black women exhibited stable or declining trends; notably, US White women showed negative drifts from ages 50 to 54 years onward. Overall (**Fig. 4B**), native Korean women had the highest increase (5.3%; 95% CI, 5.01–5.61), followed by native Japanese women (3.8%; 95% CI, 3.63–3.91) and US Korean women (2.8%; 95% CI, 1.83–3.75). US Japanese women experienced a modest increase (1.6%; 95% CI, 1.02–2.19). Comparatively, US Black women showed a small but significant increase (0.2%; 95% CI, 0.11–0.36), whereas US White women had a slight decline that was not significant (–0.06%; 95% CI, –0.16 to 0.03). Overall, APCs were significant for all populations except US White women ($P = 0.179$). In contrast, local APCs were nonsignificant for US Japanese, US Koreans, and native Koreans, whereas all other populations showed significant changes ($P < 0.001$). Detailed APC estimates with 95% CIs are provided in Supplementary Table S3.

Sensitivity analyses

Sensitivity analyses were conducted using expanded population datasets (Supplementary Table S3). US Japanese women were analyzed by combining cancer registries from Hawaii, Los Angeles, and San Francisco (Supplementary Fig. S1). The results revealed patterns consistent with those observed among Hawaiian Japanese women; however, slightly higher period effects and a greater overall net drift were observed (1.9% vs. 1.6%). In addition, analysis of the national Korean registry yielded results similar to those of the primary analysis, although the net drift was higher (5.8% vs. 5.3%; Supplementary Fig. S2).

Discussion

Distinct population-specific patterns in breast cancer incidence were revealed in this study. Native Japanese and native Korean

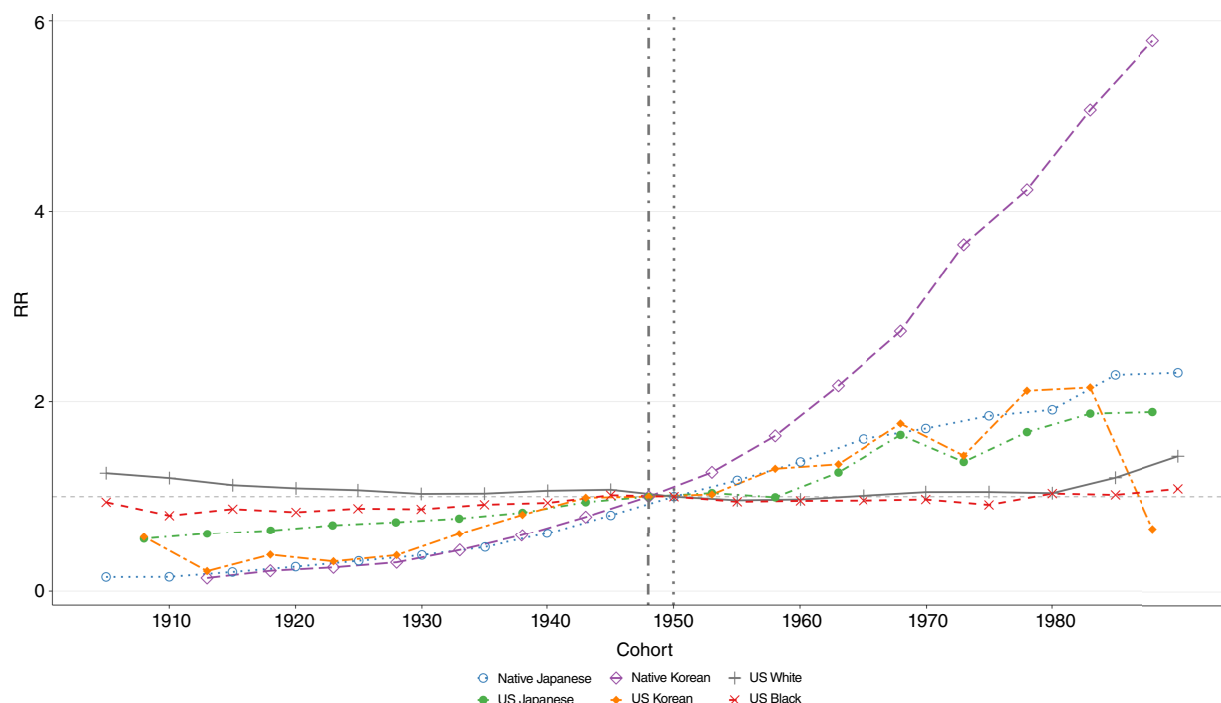


Figure 3.

Impact of cohort effects on breast cancer incidence. Cohort RRs by 5-year birth cohorts, plotted at the midpoint of each cohort interval. The faint vertical dotted line marks the reference cohort (See **Table 1** for the reference cohort used for each population).

women had the lowest overall rates, whereas US White women had the highest. The traditional age curve, characterized by an early peak around 45 to 49 years, was evident among native Japanese, native Korean, and US Korean women. In contrast, US Japanese women exhibited later peaks, resembling patterns observed in US White and

US Black women. Since the early 2000s, breast cancer incidence has increased sharply among native Japanese and Koreans, whereas trends in US populations have been modest or stable. Cohort effects indicated sustained risk increases in women born after 1950 among native Asians and US Koreans, with no consistent pattern in other

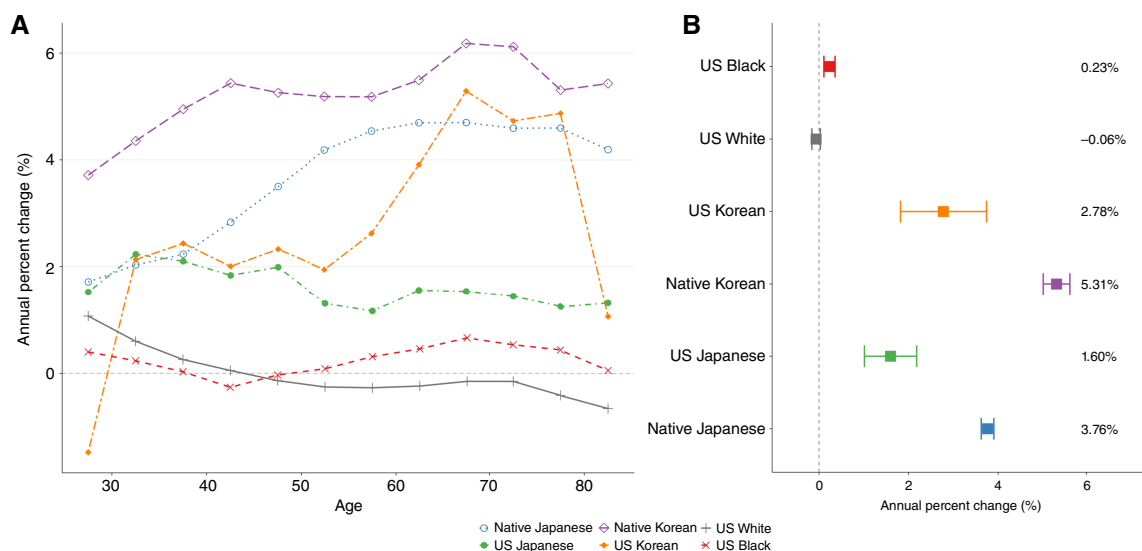


Figure 4.

Annual percent change in breast cancer incidence with 95% CIs. **A**, Age-specific annual percent change (local) by 5-year age group, plotted at the midpoint of each age interval. **B**, Overall annual percent change (net).

US groups. Annual percentage changes were greatest in native Koreans, followed by native Japanese, US Koreans, and US Japanese, whereas US Black and White women showed only slight increases and declines, respectively. These divergent trends underscore the influence of changing exposures and environments across generations and populations. Our results build on the findings of Matsuno and colleagues (11), who reported that Japanese Americans converge toward patterns of US White women whereas native Japanese women retain a midlife peak. Using extended data and APC modeling, we confirm these trends and additionally demonstrate that Korean Americans remain closer to native Koreans, highlighting heterogeneity among Asian Americans.

The contrasting age curves highlight key differences in population demographics. The cross-sectional curve, which shows observed incidence at a fixed period across age groups, confirmed the earlier peak at ages 45 to 49 years in Asian populations (native Japanese, native Koreans, and US Koreans), whereas US Japanese women showed similar results to those of US White and Black women, with a higher incidence at older ages. In contrast, US Koreans align more closely with native Asians, possibly reflecting distinct migration histories, with a large early Japanese migration wave beginning in the late 19th/early 20th century. The Japanese migration was larger than the Korean migration before 1965. Korean immigration expanded substantially after the 1965 Immigration and Nationality Act (24, 25). Consistent with this, contemporary acculturation indicators differ between groups (foreign-born share: 57% vs. 25%; English proficiency among ≥ 5 years of age: 68% vs. 87%; and US-born median age: 20.5 vs. 34.4 years for Korean vs. Japanese Americans in 2023; ref. 26) and align with evidence that breast cancer risk differs by nativity among Asian American women (27). Our cross-sectional age-incidence curves show the classic early-midlife peak among native Japanese and Koreans. This descriptive pattern is sometimes discussed under a “two-disease” model of breast cancer incidence in Asian versus Western populations, conceptualized as overlapping early- and late-onset components (28, 29). However, the longitudinal curve, which adjusts for period effects and follows a single cohort as the individuals age, showed similar increases in incidence with increasing age across all groups. The longitudinal age curve also shows an inflection around menopausal ages (Clemmesen hook), a feature that is often less apparent in cross-sectional curves when cohort effects are changing (30, 31). This suggests that the earlier peak observed in cross-sectional curves reflects generational (cohort) effects rather than biological differences (10). Although postmenopausal estrogen receptor-positive tumors are relatively less common in Asia, there are no overall significant biological differences (28). The contrasting age-incidence curves are therefore most likely driven by different constellations of risk factors acting on at least 2 etiologic components, with Asian patterns expected to shift toward Western-type profiles as urbanization and lifestyle changes progress (10, 28). Cross-sectional curves can be misleading when risk increases across cohorts, especially when increases are substantial, creating the false appearance of declining rates with age (23). These findings reinforce that breast cancer is fundamentally similar across populations, with differences driven by risk profiles and temporal exposures (10). Consequently, longitudinal curves provide a clearer depiction of lifetime risk and are more appropriate for guiding screening and prevention strategies, particularly in regions undergoing rapid epidemiologic transition (10, 23).

The pronounced period and cohort effects in native Japanese and Korean women reflect sustained increases in risk since the

1980s, with incidence nearly doubling by the 2010s. Beyond individual-level lifestyle factors, contextual drivers likely contribute to population differences in observed incidence. These period effects may reflect contextual factors, including screening program timing and uptake, diagnostic intensity and registry completeness, and broader health-system capacity differences. In the United States, mammography has been widely available since the late 1980s, and self-reported screening coverage among women 50 to 74 years of age has remained relatively high and stable (around 70%) since 2000 (32). Consequently, more recent period effects in the United States may capture only modest changes in screening intensity and diagnostic practice. In contrast, Japan introduced population-based mammography screening in 2000 (expanded to women in their 40s from 2004), and Korea initiated organized breast cancer screening through the National Cancer Screening Program in 1999 (expanded coverage began in 2002; refs. 33–35). These differences in the timing and uptake of screening are consistent with the more pronounced positive period effects observed in Japan and Korea compared with the United States. Improvements in registry completeness have also improved data accuracy, providing more reliable incidence estimates. Thus, the observed increases in incidence reflect the combined effects of actual changes in risk factors and evolving detection and registration practices. However, these sustained increases are unlikely to be attributable to detection alone or to data quality because screening participation remains lower in Japan and Korea than in the United States (33, 36). The magnitude of the increase suggests that broader determinants beyond detection and registration practices are contributing, particularly among Japanese and Korean women. These include delayed childbearing, reduced parity amid rapidly declining birth rates, earlier age at menarche, and other lifestyle changes, such as diet, physical activity, obesity, and alcohol use associated with rapid economic development and urbanization, often termed “Westernization” (6, 37–40). For example, both countries have shifted toward more energy-dense and processed diets (41–43), which meta-analyses have linked to increased breast cancer risk (44). In contrast, period effects were minimal in US populations, including Asian Americans, which is consistent with earlier and more widespread mammography uptake, declines in postmenopausal hormone therapy use, and a plateau in incidence since the early 2000s (45). Cohort trends also diverged; sharp increases were observed after the 1950 birth cohorts among native Asians and US Koreans, whereas US Whites, US Blacks, and US Japanese showed modest or no changes. These cohort patterns are consistent with previously described generational shifts in reproductive behavior and lifestyle risk factors, particularly in Japan and Korea (37, 38). Consistent with our findings, cohort-driven increases in female breast cancer incidence have also been reported in other Asian settings, including Taiwan and Thailand (46). These results are consistent with a breast cancer “transition” in terms of rapidly changing exposures and detection over time, not as an inevitable consequence of modernization. The observed increases likely reflect modifiable environmental and social conditions and therefore represent opportunities for prevention rather than a fixed trajectory.

Age-specific incidence continued to increase most sharply among native Japanese and Koreans, especially in midlife, whereas trends were flatter among US Japanese and largely stable or declining among US Whites and Blacks. Similar increases have been reported among younger and middle-aged women in Asia, as well as among

adolescents and young adults in high-income countries (47, 48). Overall, the APC analyses indicated sustained annual increases in native Japanese and native Korean women, with smaller increases among US Koreans and US Japanese; trends in US Whites were stable, and increases among US Blacks were minimal. These annual gains correspond with divergent mortality patterns. Although breast cancer–related deaths have declined significantly in the United States, partly due to widespread screening and advances in treatment, mortality continues to increase in Japan and Korea (49, 50). Sustained positive drifts in East Asian populations imply compounding annual growth, translating into a rapid escalation of population-level burden within a single generation (14, 47, 51). APC projections from Taiwan further suggest that the previously steep cohort acceleration may slow down in more recent cohorts, with a potential leveling of incidence (52). Current prevention strategies include national prevention frameworks and screening programs. In Japan, the national health promotion strategy (Health Japan 21) strengthens primary prevention through a life-course approach and sets targets to increase participation in the national cancer screening program; similarly, in Korea, the Korean National Code Against Cancer provides national prevention guidance, alongside organized early detection through the National Cancer Screening Program (53, 54). As incidence increases among younger and middle-aged cohorts, timely uptake and equitable access to these measures will become increasingly important for reducing the future disease burden.

This study applied APC modeling using the NCI APC web tool, which mitigates the identification problem inherent in APC analysis. Key study strengths include the use of standardized, high-quality, population-based registry data across multiple populations and confirmation of robustness through sensitivity analyses. This study's limitations include reliance on aggregate registry data, which precludes adjustment for individual-level risk factors, such as reproductive history or hormone use. Furthermore, the absence of detailed national screening data makes it difficult to determine whether observed increases reflect a true increase in underlying risk or reflect improved detection. Japanese data were derived from 3 predominantly nonurban regional registries. Therefore, our estimates may underrepresent the absolute breast cancer incidence in metropolitan Japan, as national cancer statistics indicate meaningful geographic variation across prefectures (2). Similar urban–rural heterogeneity has also been reported elsewhere, with Taiwan showing different incidence trends by urbanicity (31). However, the national representativeness of the registries has been confirmed, and relative age, period, and cohort patterns are likely comparable (3). US Japanese and US Korean incidence rates were derived from Hawaii and Los Angeles County registries, respectively. These estimates may not reflect all the geographic variations in these populations across the United States, although sensitivity analyses using additional data for the Japanese population showed similar patterns. Nativity information, including migration status and residency duration, was not available, which limited inference on within-group heterogeneity among US Asian populations. Furthermore, a few instances of race/ethnicity misclassification may have occurred in the Japanese registries, but these instances are unlikely to significantly affect population-level incidence patterns. Despite the

high quality of the registries included, residual differences in completeness and case ascertainment between countries and over time may contribute to observed period effects, particularly during the early years of registration. Finally, shorter follow-up or incomplete registry coverage in certain groups may have produced unstable estimates, particularly for the youngest birth cohorts. Future research should extend these population-level findings through individual-level analyses that integrate risk factors. Comparative studies of screening practices and detection bias between native and immigrant populations are also needed to clarify observed differences. A follow-up, integrated database study linking cancer registries with screening records and migration variables (e.g., country of birth, age at migration, and duration of residence) would help quantify environmental versus selection effects and enable cohort-specific forecasting.

Our findings reveal that breast cancer incidence is increasing rapidly in East Asia and among US Korean women, whereas rates have plateaued or declined in long-industrialized US populations. These patterns reflect differing stages of the epidemiologic transition and illustrate how generational shifts and migration influence risk. Population-specific strategies for early detection, primary prevention, and timely treatment are urgently needed to address the evolving burden in both native and diasporan communities.

Data Availability

All data are publicly available from the Cancer Information Service (Japan; https://ganjoho.jp/reg_stat/statistics/data/dl/en.html), CI5 (IARC; <https://ci5.iarc.fr>), and SEER 8 (<https://seer.cancer.gov>).

Authors' Disclosures

K. Katanoda reports grants from Ministry of Health, Labour and Welfare of Japan during the conduct of the study. No disclosures were reported by the other authors.

Authors' Contributions

A.K. Abubakar: Conceptualization, data curation, software, formal analysis, validation, visualization, methodology, writing—original draft, writing—review and editing. **H. Jamil:** Formal analysis, visualization, methodology, writing—review and editing. **H. Tanaka:** Conceptualization, data curation, supervision, funding acquisition, writing—review and editing. **K. Katanoda:** Conceptualization, resources, data curation, supervision, funding acquisition, validation, methodology, writing—review and editing.

Acknowledgments

The author acknowledges the use of Grammarly, a generative artificial intelligence tool, to refine sentence structure and improve clarity of the manuscript. All intellectual content, critical analysis, and interpretations remain the sole work of the authors. This study was supported by a Health Labour Sciences Research Grant of the Ministry of Health, Labour and Welfare of Japan (23EA1009) awarded to K. Katanoda.

Note

Supplementary data for this article are available at Cancer Epidemiology, Biomarkers & Prevention Online (<http://cebp.aacrjournals.org/>).

Received January 20, 2026; revised March 23, 2026; accepted April 29, 2026; posted first May 1, 2026.

References

1. Bray F, Laversanne M, Sung H, Ferlay J, Siegel RL, Soerjomataram I, et al. Global cancer statistics 2022: GLOBOCAN estimates of incidence and

mortality worldwide for 36 cancers in 185 countries. *CA Cancer J Clin* 2024;74:229–63.

2. National Cancer Center. Breast: Cancer Statistics. 2025. [cited 2025 Jul 9]. Available from: https://ganjoho.jp/reg_stat/statistics/stat/cancer/14_breast.html.
3. Katanoda K, Hori M, Saito E, Shibata A, Ito Y, Minami T, et al. Updated trends in cancer in Japan: incidence in 1985–2015 and mortality in 1958–2018—a sign of decrease in cancer incidence. *J Epidemiol* 2021;31:426–50.
4. Althuis MD, Dozier JM, Anderson WF, Devesa SS, Brinton LA. Global trends in breast cancer incidence and mortality 1973–1997. *Int J Epidemiol* 2005;34:405–12.
5. Ellington TD, Miller JW, Henley SJ, Wilson RJ, Wu M, Richardson LC. Trends in breast cancer incidence, by race, ethnicity, and age among women aged ≥ 20 years - United States, 1999–2018. *MMWR Morb Mortal Wkly Rep* 2022;71:43–7.
6. Liu R, Kitamura Y, Kitamura T, Sobue T, Sado J, Sugawara Y, et al. Reproductive and lifestyle factors related to breast cancer among Japanese women: an observational cohort study. *Medicine* 2019;98:e18315.
7. Gangnon RE, Sprague BL, Stout NK, Alagoz O, Weedon-Fekjaer H, Holford TR, et al. The contribution of mammography screening to breast cancer incidence trends in the United States: an updated age-period-cohort model. *Cancer Epidemiol Biomarkers Prev* 2015;24:905–12.
8. De P, Neutel CI, Olivetto I, Morrison H. Breast cancer incidence and hormone replacement therapy in Canada. *J Natl Cancer Inst* 2010;102:1489–95.
9. Saika K, Sobue T. Epidemiology of breast cancer in Japan and the US. *Japan Med Assoc J* 2009;52:39–44.
10. Sung H, Rosenberg PS, Chen W-Q, Hartman M, Lim W-y, Chia KS, et al. Female breast cancer incidence among Asian and Western populations: more similar than expected. *J Natl Cancer Inst* 2015;107:djv107.
11. Matsuno RK, Anderson WF, Yamamoto S, Tsukuma H, Pfeiffer RM, Kobayashi K, et al. Early- and late-onset breast cancer types among women in the United States and Japan. *Cancer Epidemiol Biomarkers Prev* 2007;16:1437–42.
12. Suzuki S, Kojima M, Tokudome S, Mori M, Sakauchi F, Wakai K, et al. Obesity/weight gain and breast cancer risk: findings from the Japan collaborative cohort study for the evaluation of cancer risk. *J Epidemiol* 2013;23:139–45.
13. Anderson WF, Matsuno RK, Sherman ME, Lissowska J, Gail MH, Brinton LA, et al. Estimating age-specific breast cancer risks: a descriptive tool to identify age interactions. *Cancer Causes Control* 2007;18:439–47.
14. Nguyen PT, Hori M, Matsuda T, Katanoda K. Cancer prevalence projections in Japan and decomposition analysis of changes in cancer burden, 2020–2050: a statistical modeling study. *Cancer Epidemiol Biomarkers Prev* 2023;32:1756–70.
15. Ziegler RG, Hoover RN, Pike MC, Hildesheim A, Nomura AM, West DW, et al. Migration patterns and breast cancer risk in Asian-American women. *J Natl Cancer Inst* 1993;85:1819–27.
16. Rosenberg PS, Anderson WF. Age-period-cohort models in cancer surveillance research: ready for prime time? *Cancer Epidemiol Biomarkers Prev* 2011;20:1263–8.
17. Minami Y, Tsubono Y, Nishino Y, Ohuchi N, Shibuya D, Hisamichi S. The increase of female breast cancer incidence in Japan: emergence of birth cohort effect. *Int J Cancer* 2004;108:901–6.
18. Ito Y, Ioka A, Nakayama T, Tsukuma H, Nakamura T. Comparison of trends in cancer incidence and mortality in Osaka, Japan, using an age-period-cohort model. *Asian Pac J Cancer Prev* 2011;12:879–88.
19. Wu B, Li Y, Shi B, Zhang X, Lai Y, Cui F, et al. Temporal trends of breast cancer burden in the Western Pacific Region from 1990 to 2044: implications from the Global Burden of Disease Study 2019. *J Adv Res* 2024;59:189–99.
20. Wang F, Liu S, Li J, Shi Y, Geng Z, Ji Y, et al. Burdens of breast cancer and projections for 2030 among women in Asia: findings from the 2021 Global Burden of Disease Study. *Curr Oncol* 2025;32:267.
21. Deng T, Zi H, Guo X-P, Luo L-S, Yang Y-L, Hou J-X, et al. Global, regional, and national burden of breast cancer, 1990–2021, and projections to 2050: a systematic analysis of the Global Burden of Disease Study 2021. *Thorac Cancer* 2025;16:e70052.
22. Du XL, Song L. Breast cancer incidence trends in Asian women aged 20 or older as compared to other ethnic women in the United States from 2000 to 2018 by time period, age and tumor stage. *Cancer Epidemiol* 2022;76:102076.
23. Rosenberg PS, Check DP, Anderson WF. A web tool for age-period-cohort analysis of cancer incidence and mortality rates. *Cancer Epidemiol Biomarkers Prev* 2014;23:2296–302.
24. Bankston CL III. Immigration and Nationality Act of 1965. 2012 [cited 2026 Feb 25]. Available from: <https://immigrationtounitedstates.org/594-immigration-and-nationality-act-of-1965.html>.
25. Gregory JN. Asian American and Pacific Islander migrations: history and geography. University of Washington; 2022 [cited 2026 Apr 28]. Available from: https://depts.washington.edu/moving1/asian_migration.shtml.
26. Pew Research Center. Asian Americans - research and data from Pew Research Center. Washington (DC): Pew Research Center; 2025 [cited 2025 Aug 17]. Available from: <https://www.pewresearch.org/topic/race-ethnicity/racial-ethnic-groups/asian-americans/>.
27. Morey BN, Gee GC, von Ehrenstein OS, Shariff-Marco S, Canchola AJ, Yang J, et al. Higher breast cancer risk among immigrant Asian American women than among US-born Asian American women. *Prev Chronic Dis* 2019;16:E20.
28. Leong SPL, Shen Z-Z, Liu T-J, Agarwal G, Tajima T, Paik N-S, et al. Is breast cancer the same disease in Asian and Western countries? *World J Surg* 2010;34:2308–24.
29. Lin C-H, Yap YS, Lee K-H, Im S-A, Naito Y, Yeo W, et al. Contrasting epidemiology and clinicopathology of female breast cancer in Asians vs the US population. *J Natl Cancer Inst* 2019;111:1298–306.
30. Clemmesen J. Carcinoma of the breast; results from statistical research. *Br J Radiol* 1948;21:583–90.
31. Lee PJ, Huang JR, Chen YC, Su SY, Chiang CJ, Yang YW, et al. Urban-rural disparity in birth cohort effects on breast cancer incidence. *J Urban Health* 2023;100:341–54.
32. White A, Thompson TD, White MC, Sabatino SA, de Moor J, Doria-Rose PV, et al. Cancer screening test use - United States, 2015. *MMWR Morb Mortal Wkly Rep* 2017;66:201–6.
33. Chotai N, Renganathan R, Uematsu T, Wang J, Zhu Q, Rahmat K, et al. Breast cancer screening in Asian countries: epidemiology, screening practices, outcomes, challenges, and future directions. *Korean J Radiol* 2025;26:743–58.
34. Hamashima C; Japanese Research Group for the Development of Breast Cancer Screening Guidelines; Hamashima CC, Hattori M, Honjo S, Kasahara Y, et al. The Japanese guidelines for breast cancer screening. *Jpn J Clin Oncol* 2016;46:482–92.
35. Lim SM, Lee HY, Choi KS, Jun JK, Park EC, Kim Y, et al. Trends of mammography use in a national breast cancer screening program, 2004–2008. *Cancer Res Treat* 2010;42:199–202.
36. Abubakar AK, Kaneda Y, Ozaki A, Saito H, Murakami M, Hori D, et al. Two-year-span breast cancer screening uptake in Japan after the COVID-19 pandemic and its association with the COVID-19 vaccination. *Cancers* 2024;16:1783.
37. Iwasaki M, Tsugane S. Risk factors for breast cancer: epidemiological evidence from Japanese studies. *Cancer Sci* 2011;102:1607–14.
38. Yoo K-Y, Kim Y, Park SK, Kang D. Lifestyle, genetic susceptibility and future trends of breast cancer in Korea. *Asian Pac J Cancer Prev* 2006;7:679–82.
39. Ang BH, Teo S-H, Ho W-K. Systematic review and meta-analysis of lifestyle and reproductive factors associated with risk of breast cancer in Asian women. *Cancer Epidemiol Biomarkers Prev* 2024;33:1273–85.
40. Iwase M, Taniyama Y, Koyanagi YN, Kasugai Y, Oze I, Masuda N, et al. A century of change: unraveling the impact of socioeconomic/historical milestones on age at menarche and other female reproductive factors in Japan. *J Epidemiol* 2024;34:387–92.
41. Koivai K, Takemi Y, Hayashi F, Ogata H, Matsumoto S, Ozawa K, et al. Consumption of ultra-processed foods decreases the quality of the overall diet of middle-aged Japanese adults. *Public Health Nutr* 2019;22:2999–3008.
42. Lee H, Yim Y, Cho Y, Oh J, Kim S, Son Y, et al. Long-term trends and patterns in ultra-processed food consumption among Korean adults from 1998 to 2022. *Sci Rep* 2025;15:4131.
43. Song S, Shim JE, Song WO. Trends in total fat and fatty acid intakes and chronic health conditions in Korean adults over 2007–2015. *Public Health Nutr* 2019;22:1341–50.
44. Karimi M, Rabiee R, Hooshmand F, Aghapour B, Ahmadzadeh M, Havaei E, et al. Consumption of fast foods and ultra-processed foods and breast cancer risk: a systematic review and meta-analysis. *Glob Health Res Pol* 2025;10:25.
45. Centers for Disease Control and Prevention (CDC). Decline in breast cancer incidence-United States, 1999–2003. *MMWR Morb Mortal Wkly Rep* 2007;56:549–53.

46. Chen Y-C, Lien W-C, Su S-Y, Jhuang J-R, Chiang C-J, Yang Y-W, et al. Birth cohort effects in breast cancer incidence: global patterns and trends. *Am J Epidemiol* 2022;191:1990–2001.
47. Li Y, Zheng J, Deng Y, Deng X, Lou W, Wei B, et al. Global burden of female breast cancer: age-period-cohort analysis of incidence trends from 1990 to 2019 and forecasts for 2035. *Front Oncol* 2022;12: 891824.
48. di Martino E, Smith L, Bradley SH, Hemphill S, Wright J, Renzi C, et al. Incidence trends for twelve cancers in younger adults—a rapid review. *Br J Cancer* 2022;126:1374–86.
49. Katanoda K, Tanaka H, Ito Y. International trends in cancer mortality revisited: Japan's site-specific progress and remaining challenges until 2024. *Jpn J Clin Oncol* 2026 Mar 5 [Epub ahead of print].
50. Berry DA, Cronin KA, Plevritis SK, Fryback DG, Clarke L, Zelen M, et al. Effect of screening and adjuvant therapy on mortality from breast cancer. *N Engl J Med* 2005;353:1784–92.
51. Liao L. Inequality in breast cancer: global statistics from 2022 to 2050. *Breast* 2025;79:103851.
52. Chen Y-C, Su S-Y, Jhuang J-R, Chiang C-J, Yang Y-W, Wu C-C, et al. Forecast of a future leveling of the incidence trends of female breast cancer in Taiwan: an age-period-cohort analysis. *Sci Rep* 2022;12:12481.
53. Wada Y. Start of the 12-year initiative for the third term of Health Japan 21. *Health Med Sci* 2024;73:68–78.
54. Choi Y, Oh J-K, Byeon A, Kim B. The Korean National Codes Against Cancer: background of their establishment and the revision process. *Epidemiol Health* 2025;47:e2025027.